

Peer Review Filter Teacher Notes

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	Time	30-40 min
Difficulty	Medium	ID	peer-review-filter

Big idea: Scientific knowledge is filtered by community criticism, not just produced by individual discovery.

Overview

Students act as reviewers for a fictional study and decide whether flaws require revision, rejection, or cautious acceptance.

Student Prompt

Inspect Peer Review Filter Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Peer Review Filter Stimulus A	Student-facing stimulus 1: Study claim: a lavender scent improves memory scores by 18% in one class of 18 students. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Peer Review Filter Stimulus B	Student-facing stimulus 2: Flaws: no random assignment, small sample, unclear scoring, but transparent method and plausible follow-up. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Peer Review Filter Reveal / Twist	Student-facing stimulus 3: Flaws: no random assignment, small sample, unclear scoring, but transparent method and plausible follow-up. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Scientific knowledge is filtered by community criticism, not just produced by individual discovery.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Peer Review Filter Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Peer Review Filter Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	How does criticism strengthen knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how scientific knowledge is filtered by community criticism, not just produced by individual discovery.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Write the fictional study so it has strengths as well as flaws.
- Give reviewers different priorities to simulate real disagreement.

Ready-to-Use Examples

- Study claim: a lavender scent improves memory scores by 18% in one class of 18 students.
- Flaws: no random assignment, small sample, unclear scoring, but transparent method and plausible follow-up.

Suggested Timing

Stage	Teacher move	Watch for
1	Give groups a fictional study with several strengths and weaknesses.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Students annotate method, sample, measurement, claim size, and limitations.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Groups write reviewer comments and a decision: accept, revise, or reject.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Compare reviewer decisions and identify criteria behind disagreement.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief peer review as a social process for improving knowledge.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Peer review is not a truth machine; it is organized criticism.
- Which flaw changes the conclusion, and which flaw can be repaired?

Common Misconceptions

- Students may think observation happens before theory in a simple linear order.
- Students may overclaim from a classroom model without naming its limits.

Assessment Look-Fors

- Names variables, assumptions, controls, anomalies, and limits.
- Explains how methods shape what can count as scientific knowledge.

Differentiation and Adaptations

- Support: provide three marked flaws.
- Challenge: include conflicting reviewer recommendations and require an editor decision.

Extension Tasks

- Students rewrite the study conclusion to match the evidence more responsibly.
- Compare peer review with fact-checking, editing, or legal cross-examination.

Related Activities

- Placebo Trial Design
- Replication Debate
- Observation is Theory-Laden